Active Maintenance Report: coppock_2014

Coppock (2014, Journal of Experimental Political Science): Information Spillovers, and its 2016 corrigendum

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This repository holds the actively maintained replication code for Coppock (2014) and for the corrigendum that corrects it, together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/xps.2014.9
Corrigendum	10.1017/xps.2016.8
Replication archive	10.7910/DVN/28664
Reanalyzed experiment	Butler and Nickerson (2011), 10.1561/100.00011019

The corrigendum is a separately published article with its own DOI, its own year and its own pagination, so it is cited here in its own right rather than as an attachment to the 2014 paper. It matters for what follows: the deposited archive was never updated after it appeared, so the code at the DOI above still produces the results the corrigendum retracts.

The data are not redistributed here. The deposit is 3.0 MB across 30 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published figure or table, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `maintained/in_text_claims.R` prints every claim the published record makes, each beside the sentence that makes it. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the extraction of every number the article, its appendix and the corrigendum print, and `build_ground_truth.R` assembles the comparison and runs the coverage gate. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_2014.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 30 files, and produces every figure and table into `maintained/output/`. Required packages: tidyverse, sf, spdep, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about five minutes, most of it in the four-parameter appendix model. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

What this repository certifies is the corrigendum, not the article. The 2016 corrigendum retracts the spillover conclusion of the 2014 paper and prints corrected estimates in their place. Those are the numbers this rewrite stands behind.

Coverage is a separate question from certification, and the ground truth covers the whole published record: every number the article prints, every cell of its two appendix tables, and every number in the corrigendum. Where the corrigendum supersedes an article value, the row records both and says so. Two of the findings below sit entirely outside the corrigendum's scope and would have been invisible to a check organized around the corrected figures.

1.1 Does the deposited archive run?

Not as deposited, and the reason is structural rather than package rot. Every figure script opens with `load("Replication Archive/CoppockJEPS.rdata")` and writes into `Replication Archive/figures/`. No such directory is in the deposit: all 30 files sit flat at the top level, so the folder structure the code expects did not survive upload. A reader has to build three directories before any script runs. That is a five-minute repair and it is invisible until you attempt it.

Once the directories exist the figure scripts run. The data preparation script that builds their input does not: it calls `maptools`, which left CRAN in 2023.

What the archive does not contain is any trace of the correction. The deposited data still holds the corrupted similarity matrix the corrigendum describes, so the figures a reader produces from it today are the ones the corrigendum retracts.

1.2 Does the maintained rewrite reproduce the corrigendum?

Yes, and both of its panels rather than only the corrected one. 3 of the 5 corrected values return exactly, and the two that sit one grid step away do so for a reason recorded below and not because the analysis differs. The six values the corrigendum restates from the original analysis return exactly as well: its four confidence bounds once that analysis is run on the ideal

points the corrigendum used rather than the deposited ones, and its two point estimates on the article's own surface, which is what its sentence describes. That last point took a while to establish, and it reverses a reading this repository carried earlier; the section "The corrigendum" below sets out both.

Getting there required two things that are worth stating, because either one alone leaves the corrigendum looking irreproducible.

The ideal points are not the deposited ones. Similarity between legislators is measured from W-NOMINATE scores estimated on the previous session's roll calls. The archive behind the corrigendum carries different scores from the deposited archive: 68 of the 70 differ, by up to 0.533, correlation 0.978. The transform that turns scores into a similarity matrix is byte-identical in both, `(2 - |x - y|)/2` with a zeroed diagonal, so the matrices differ only because their inputs do.

The cause is a package. Both archives call `wnominate(rollcalls, polarity = c(1, 2), minvotes = 10)`, identically. The deposited archive loads the package with no version constraint; the corrigendum archive pins it, with a comment reading "be sure to use version 0.98" and a `stopifnot` to enforce it. `wnominate` 0.98 was current when the analysis was written and 0.99 arrived in September 2014, between the analysis and publication. Running today's `wnominate` 1.5 on the deposited roll calls puts the result closer to the corrigendum's scores (r = 0.993) than to the deposited ones (r = 0.976), which is the signature of an estimator that moved. The deposited `datapreparation.R` can no longer regenerate the deposited scores, and nothing in it says so.

The 70 scores are in `recovered/`, committed because they exist nowhere public.

The analysis predates R 3.6. The p-value surface draws 685,000 times, and R 3.6.0 replaced the algorithm behind `sample()`. Under the sampler of the day, with the corrected scores, the surface this repository produces is identical to the one saved in the corrigendum's own archive: all 1,369 cells, maximum difference exactly zero.

Those are two different questions and they are kept apart. `text_sampler_reproducibility.R` re-runs both of the corrigendum's panels under `RNGkind(sample.kind = "Rounding")`, because whether the archive reproduces is a question about the archive. The rewrite itself keeps the current sampler, because it is the code we would write now. The two remaining differences are one grid step each and are recorded as belonging to the environment. The sampler matters to the corrected panel and not to the original one, which returns the same four bounds either way, and that asymmetry is what rules the sampler out as the explanation of the discrepancy taken up below.

One published value is not identified rather than unreproduced. The confidence region reaches 0.2, the top of the grid, in the corrigendum's surface as well as this one, so the printed 0.175 is the last interior grid point rather than a bound the data support.

1.3 What this repository does not certify

The archive's figure scripts produce the article's p-value maps, and this rewrite reproduces them bit for bit where they can be compared. The estimates read off those maps are in the ground truth, but as a record of what the deposit does rather than as results anyone should now use: the corrigendum supersedes the two figures that carry the coding error, and the rewrite reads its point estimates by the method the article describes rather than the one its code implements, which moves three of them.

Three coding defects in the deposit are worth recording even so, since none is mentioned by the corrigendum. `CoppockJEPS_appendixfigure4.R` never calls `set.seed`, so Appendix Figure 4 differs on every run and cannot be reproduced by anyone. Five scripts read a point estimate off the first and second entries of a plateau index matrix in column-major order, which indexes the indirect axis with a position from the direct one; `figure2code.R` takes the median, which is what the paper describes. `appendixfigure5.R` draws its assignment vector once outside the simulation loop, so all fifty draws are the same assignment and every p-value is 0 or 1, which is visible in the published figure as solid blocks.

2 Paper overview

Citation: Coppock, A. (2014). "Information spillovers: Another look at experimental estimates of legislator responsiveness." *Journal of Experimental Political Science*, 1(2), 159-169. DOI: 10.1017/xps.2014.9

Corrigendum: Coppock, A. (2016). *Journal of Experimental Political Science*, 3(2), 206-208. DOI: 10.1017/xps.2016.8

Summary: Butler and Nickerson (2011) mailed district-level public opinion data to 35 of the 70 members of the New Mexico House before a 2008 vote on a tax rebate bill, in a matched-pair randomization, and estimated the effect of that information on roll call votes. The reanalysis asks what happens when the non-interference assumption is dropped, on the grounds that legislatures are information-sharing networks. Legislators are assumed to be exposed to treatment through their ideologically similar colleagues, with similarity measured from W-NOMINATE ideal points estimated on the preceding session's roll calls. Inference follows Bowers, Fredrickson and Panagopoulos (2013): each pair of hypothesized direct and indirect effects implies a uniformity trial, the hypothesis is re-imposed under draws from the 10,000 admissible randomizations, and the p-value is the share of draws whose sum of squared residuals the observed one exceeds. The article concluded that the total causal effect of treatment was about twice the difference in means. The corrigendum retracts the spillover part of that conclusion.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
CoppockJEPS_10000Randomizations.R	Runs; regenerates a matrix the deposit already carries, under a seed R has since changed	None needed: Z_block is inside the deposited CoppockJEPS.RData
CoppockJEPS_datapreparation.R	Fails: maptools is off CRAN, and it writes CoppockJEPS.rdata over the deposited CoppockJEPS.RData	sf::st_read plus spdep::poly2nb rebuilds the one matrix maptools was used for
CoppockJEPS_figure1acode.R	Fails on a missing directory; runs once it exists	Create Replication Archive/figures and .../modeloutputs; copy CoppockJEPS.RData in
CoppockJEPS_figure1bcode.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_figure2code.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_figure3code.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure2.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure3.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure4.R	Fails on a missing directory; runs once it exists, and sets no seed	As above; set.seed(343), as every other script does
CoppockJEPS_appendixfigure5.R	Fails on a missing directory; runs once it exists, and its simulation never permutes	As above; draw a fresh assignment per simulation
CoppockJEPS_appendixtable.R	Fails: loads seven files by bare name and writes to a hardcoded /Users/Alex path	Run from the modeloutputs directory; redirect the output

3.1 The directory that is not in the deposit

The scripts and the deposit disagree about where files live, in both directions. Every figure script reads `Replication Archive/CoppockJEPS.rdata` and writes to `Replication Archive/figures/`; the data preparation script reads `CoppockJEPSshapefile/hd_court_ordered.shp`. Neither path exists at Dataverse, where the shapefile's three components sit at the top level alongside everything else. The published appendix documents the shapefile directory as a directory, so this is an upload that flattened a tree rather than an author who never had one.

Two smaller path defects come with it, and both are invisible on macOS.

`CoppockJEPS_10000Randomizations.R` writes `CoppockJEPS_10000randomizations.rdata`, with a lower-case r, and `CoppockJEPS_datapreparation.R` loads `CoppockJEPS_10000Randomizations.rdata`, with an upper-case one. A case-insensitive filesystem resolves that; Linux does not.

More seriously, `CoppockJEPS_datapreparation.R` ends by writing `CoppockJEPS.rdata` into the working directory, and the deposited file is `CoppockJEPS.RData`. Running the data preparation script inside a copy of the archive on macOS therefore silently overwrites a deposited file with a regenerated one. That is why everything in this report was run from a scratch copy, and why `download_original.R` reports anything in `original/` that the manifest does not list.

4 The corrigendum

The corrigendum is the substance of the case, so it is worth being precise about which object is wrong.

`CoppockJEPS.RData` stores four similarity matrices. Three of them, the rank, squared-distance and geographic ones, rebuild from the deposited ideal points and shapefile exactly. The fourth, `similarity.matrix.dw`, is 1 in every off-diagonal cell and 0 on the diagonal. It asserts that every pair of legislators in the New Mexico House is maximally and identically close in ideological space, which is precisely what the corrigendum says went wrong.

The corruption is confined to that one object. The net exposure variable the published regressions use, `s.difs.dw`, was computed from a correct similarity matrix: rebuilding it from Equation (1) reproduces the deposited values to fourteen decimal places. The published analysis therefore regressed the observed outcome on a correctly built exposure measure while simulating the null distribution with the all-ones matrix. Under that matrix every unit's raw exposure is the number of treated colleagues, which is the same for everyone, so the simulated exposure carries no variation attributable to the random assignment at all. The reference distribution is far too narrow, and the confidence regions that come out of it are far too tight. That is the mechanism behind the corrigendum.

Table 2: The two panels of the corrigendum's Figure 1 as it states them and as the maintained rewrite produces them. The original panel is the analysis the article published, re-run on the ideal points the corrigendum used; the corrected panel is the correction it substitutes.

Panel	Quantity	Corrigendum	Rewrite
Original	indirect effect	-0.225	-0.225
Original	direct effect	-0.300	-0.3
Original	indirect 95% CI lower	-0.325	-0.325
Original	indirect 95% CI upper	-0.025	-0.025
Original	direct 95% CI lower	-0.575	-0.575
Original	direct 95% CI upper	0.000	0
Corrected	indirect effect	-0.150	-0.15
Corrected	direct effect	-0.250	-0.25
Corrected	indirect 95% CI lower	-0.325	-0.35
Corrected	indirect 95% CI upper	0.200	0.175
Corrected	direct 95% CI lower	-0.625	-0.625
Corrected	direct 95% CI upper	0.175	

Three things in that table are worth separating.

The corrigendum's restatement of the original result describes its own redrawn panel, and both documents are right. The corrigendum reports the original 95% confidence intervals as running from -0.325 to -0.025 for the indirect effect and from -0.575 to 0.000 for

the direct effect. Appendix Table 1 of the 2014 article gives [-0.3, -0.1] and [-0.5, -0.075]. The two disagree, and the reason is the ideal points rather than an error in either.

The uncorrected analysis is not independent of the ideal points, which is not obvious and is why the disagreement looked for a while like a defect. Under the all-ones matrix every unit's raw exposure is the number of treated colleagues, the same for everyone, so the simulated exposure varies only through the expected-exposure term of Equation (5), and that term is built from the similarity matrix the scores define. Run on the deposited scores the uncorrected model returns the appendix's intervals at the precision the appendix prints; run on the scores the corrigendum was computed from it returns the corrigendum's, all four bounds exactly. Each document reports what its own materials produce, and the sampler is not what separates them: unlike the corrected panel, this surface gives the same four bounds under the sampler of 2016 and the current one alike. The point estimates the corrigendum restates are the article's own, and they attain the highest p-value on both surfaces, so no number in that sentence is unsupported.

The corrigendum's printed figure says the same thing independently. Its "Original" panel is not the map the 2014 article printed: the article's is a rounded blob stopping well short of the vertical zero line, and the corrigendum's is wider, tilted along the diagonal, and reaches almost to zero on the direct axis. Both maps are drawn under Figure verification below and differ in exactly that way. The difference is a plateau of 90 tied cells against one of 28. The corrigendum redrew the original analysis rather than reprinting it.

The wider plateau is also why the median over it no longer returns the pair the corrigendum names. That pair is still at the top of the corrigendum's own surface, with a p-value of 1, alongside 89 other cells tied with it.

All three surfaces are computed here. The article's Figure 2 and the corrigendum's two panels are separate objects in `maintained/output/`, and no number is transcribed from either document.

The correction reproduces exactly, once two things are supplied that the deposit does not carry. Rebuilding the ideological similarity matrix from the deposited ideal points and rerunning does what the corrigendum says qualitatively, widening both intervals across zero, but it returns neither the corrected point estimates nor the corrected intervals. Seeds, simulation counts, grid width and the matrix diagonal were each varied and none of them closes the gap.

The gap closes on the ideal points. The corrigendum was computed from a different set of W-NOMINATE scores than the deposit carries, and the difference traces to an unpinned package, as the Summary sets out. With those scores, and with the sampler of the day restored, the p-value surface this repository produces is identical to the one saved in the corrigendum's own working archive: all 1,369 cells, maximum difference exactly zero.

The archive nonetheless records none of that. It was released in 2015 and never revised, so a reader who downloads it today gets the retracted figures, the scores that no longer

regenerate, and no indication that either is the case. A corrigendum that corrects a published analysis is itself a published analysis, and this one was never deposited.

The corrigendum prints two sets of values: the corrected estimates it substitutes, and a restatement of the original ones it is correcting. Both are recorded.

Of the corrected set, 3 of 5 return exactly. Of the restatement, all six return exactly, once each is laid against the panel it describes. The corrigendum gives the original direct interval as -0.575 to 0.000 where the 2014 appendix gives a narrower one, and this repository reproduces both.

5 The extraction and the two instruments

Two things stand between a published sentence and a silent error, and they are separate files that reach the same number by separate routes.

The extraction. `ground_truth/published_claims.csv` records every numeric token in the article, its online appendix and the corrigendum, together with the numbers those documents spell out in words. It has 140 rows, classified by hand as 15 definitional, 25 descriptive, 77 pipeline, 23 structural. A `pipeline` claim is an estimate the code produces; a `descriptive` claim asserts a shape, a sign or a comparison rather than a value; `structural` and `definitional` claims are design facts and properties of the formulas. The extraction is what makes coverage a question with an answer: 115 of the 140 rows are claims the pipeline can reach, and each must be printed by a claim block.

The ground truth. `ground_truth/coppock_2014_ground_truth.csv` has 116 rows, each pairing what a document prints against what `maintained/output/` produces. 89 carry a value comparison, of which 72 agree at the precision the page prints and 17 do not; 25 are descriptive claims carrying a computed truth value instead, and all 25 hold. It is generated by `ground_truth/build_ground_truth.R`, which reads every rewrite value out of `maintained/output/` rather than transcribing it.

The second instrument. `maintained/in_text_claims.R` carries each sentence verbatim in a block comment and recomputes its number from the same committed output, reaching it by its own path. Where the ground truth reads a cell of appendix Table 1 out of `table_1_constant_effects.csv`, the claims file reads the same cell out of the figure script's own CSV. It never reads the ground truth.

The gate. `build_ground_truth.R` runs the claims file as a program, under `capture.output` and in its own environment, and stops the build unless the number of printed claims equals the number of extraction rows that require one, the two sets of identifiers are equal in both directions, every printed value equals the ground truth's at the same precision, every extracted transcription equals the ground truth's, and every row carrying an adverse verdict from either

instrument names where the fault lies. A block that errors, prints nothing, or quietly reads an empty selection fails it.

There is no errata document. Every value in this article that the data do not support is already corrected by a published corrigendum, which carries its own DOI and its own citation, so a note here would duplicate the published record rather than add to it. The one exception is recorded below and has not been written up unilaterally.

5.1 What the extraction found that the ground truth had not

Table 3: Four claims outside the corrigendum's scope, none of them reachable by a check organized around published floats.

Claim	Published	Here	Where
2^35 possible random assignments, called 30 trillion two sentences later	30	0.03436	paper_internal
Figure 1b lower-support estimate in the text, against -0.375 in appendix Table 2	-0.37	-0.375	matches
Figure 3a direct effect, read as the median over the plateau the article describes	-0.550	-0.475	archive
Appendix Table 1 geographic indirect effect, from the one unseeded figure script	-0.025	0.025	archive

The first is the only one that could have warranted a correction of its own. The article writes the number of possible matched-pair random assignments twice: once as $2^{35} = 3.4 \times 10^{10}$, which the deposit confirms, and once, two sentences later, as "the 30 trillion random assignments". Thirty trillion is 3×10^{13}, three orders of magnitude above the count the same paragraph gives. Nothing turns on it, no estimate moves, and the corrigendum does not mention it. It is recorded here and deliberately not issued as an erratum: this article already carries a published corrigendum, and a second correction document for a single stray number in a paragraph that states the right figure two sentences earlier would clutter the record more than it clarifies it.

The second is not an error. Appendix Table 2 gives -0.375 and the pipeline reproduces it; the main text prints -0.37, which is the same number truncated rather than rounded, and at two decimals the two sit exactly on the rounding boundary.

The third and fourth are properties of the deposited code rather than of the article. `CoppockJEPS_figure3code.R` takes its point estimate from the first and second entries of the plateau index matrix in column-major order, so where the plateau spans more than one cell it reads a row position off the direct axis and uses it to index the indirect one; Figure 3's lower-support plateau spans 77 cells. `CoppockJEPS_appendixfigure4.R` never calls `set.seed`, so the sign the appendix prints for the geographic placebo's indirect effect is a property of one draw.

6 Number-by-number comparison

Table 4: The ground truth by document. Every published number in both is covered; the full table is the committed ground truth CSV.

Document	Rows	Compared	Agreeing	Descriptive	Holding
Article and appendix (2014)	100	78	63	21	21
Corrigendum (2016)	16	11	9	4	4

`value_script`, the column recording what the deposited scripts print, is empty throughout. This repository has no portable `extract_archive_values.R`, so the rule that the archive column be generated rather than typed is unmet here, and an empty column is a more honest statement of that than a transcribed one. What the deposited code produces is recorded instead in the section below, where the rewrite's p-value surfaces are compared with the archive's cell by cell.

6.1 Coverage float by float

A float with one row in the ground truth is not a covered float, so the table below states the fraction. The p-value maps print no numbers on their faces beyond axis ticks chosen by the plotting device, so what each one asserts is the estimate and confidence region read off it, which the article states in prose, in appendix Table 1 or 2, or in both.

Table 5: Coverage of every float in the article, its appendix and the corrigendum.

Float	Claims covered	Notes
Figure 1a	4 of 4	
Figure 1b	3 of 3	
Figure 2	5 of 5	
Figure 3	9 of 9	
Appendix Figure 1	no stated values	Prints no numbers; the 68 ideal points it plots are counted instead
Appendix Figure 2	7 of 7	
Appendix Figure 3	8 of 8	
Appendix Figure 4	8 of 8	
Appendix Figure 5	2 of 2	Prints no numbers and has no summary table; covered by the parameter count and the comparison with Figure 3
Appendix Table 1	27 of 27	
Appendix Table 2	18 of 18	
Corrigendum Figure 1	12 of 12	
Corrigendum Figure 2	no stated values	The corrigendum states no values for its corrected Figure 3, so there is nothing to compare against

Table 6: Every row where a published value and this repository disagree, with the cause.

Location	Quantity	Published	Here	Where the fault lies
Text, p. 162	The same set of random assignments, in units of 10^12	30	0.03436	paper_internal
Text, p. 166	Figure 3a direct effect, lower-support districts	-0.550	-0.475	archive
Text, p. 166	Figure 3a indirect effect, lower-support districts	-0.125	-0.175	archive
Text, p. 166	Figure 3b direct effect, higher-support districts	-0.175	-0.1875	archive
Appendix Table 1	Indirect 95% CI, rank_distance, bound 2	0.175	0.15	unresolved
Appendix Table 1	Indirect 95% CI, squared_distance, bound 1	-0.35	-0.325	unresolved
Appendix Table 1	Direct 95% CI, geographic_contiguity, bound 1	-0.38	-0.37	archive
Appendix Table 1	Direct 95% CI, geographic_contiguity, bound 2	0.08	0.09	archive
Appendix Table 1	Indirect Effect, geographic_contiguity	-0.025	0.025	archive
Appendix Table 1	Indirect 95% CI, geographic_contiguity, bound 2	0.145	0.15	archive
Appendix Table 2	Direct Effect, High Support, absolute_distance_uncorrected	-0.175	-0.1875	archive
Appendix Table 2	Direct 95% CI, High Support, absolute_distance_uncorrected, bound 1	-0.45	-0.4	unresolved
Appendix Table 2	Direct Effect, Low Support, absolute_distance_uncorrected	-0.55	-0.475	archive
Appendix Table 2	Direct 95% CI, Low Support, absolute_distance_uncorrected, bound 1	-0.825	-0.8	unresolved
Appendix Table 2	Indirect Effect, Low Support, absolute_distance_uncorrected	-0.125	-0.175	archive
Corrigendum Figure 1, corrected panel	indirect 95% CI lower	-0.325	-0.35	environment
Corrigendum Figure 1, corrected panel	indirect 95% CI upper	0.200	0.175	environment
Corrigendum Figure 1, corrected panel	direct 95% CI upper	0.175	not identified	paper_internal

7 Maintained rewrite

The rewrite lives in `maintained/`: a cleaning script and a helpers file, nine figure scripts, two table scripts, two scripts for the in-text quantities, the sampler reproducibility check, and `in_text_claims.R`.

7.1 Architecture

`clean_data.R` is the load-bearing one. It reads the deposited `CoppockJEPS.RData`, rebuilds all four similarity matrices from the deposited ideal points and shapefile, and prints how far each rebuilt matrix sits from the deposited one, which is what localizes the defect: three deviations of zero and one of 1. It then computes expected and observed net exposures for each matrix, and keeps the deposited all-ones matrix alongside the corrected one so that the uncorrected figures can be reproduced rather than described.

Every spillover figure is therefore computed twice, once from each matrix, and both versions go into `output/`. Figure 2 is computed three times, because the corrigendum's original panel is a third surface: the published all-ones matrix on the corrigendum archive's ideal points. Nothing is transcribed: the corrigendum's numbers appear in the ground truth as comparison targets and nowhere in the code.

The p-value grid itself is one function, `pvalue_grid()` in `helpers.R`, taking the similarity matrix, the expected exposures and the observed exposure as separate arguments. That separation is deliberate. The difference between the published analysis and the corrected one is exactly which matrix goes into the simulation while the observed regressor stays the same, and a helper that fixed either of them internally would hide the only choice in the paper that matters.

The rewrite's p-value matrices were checked cell by cell against the ones the deposited scripts save. Six of the seven are bit-identical: Figure 1a's 71 cells, Figure 1b's 2,205, Figure 2's 1,369, Figure 3's 1,305, Appendix Figure 2's 1,681 and Appendix Figure 3's 2,401, with a maximum absolute deviation of exactly zero in every case. Appendix Figure 4 is the exception, and only because its script sets no seed. Agreement at that level is what makes the point-estimate differences interpretable: the surfaces are the same, so any difference in what is read off them is a difference in the reading.

One defect that comparison could not catch, and a published figure did. `tile_grid()` turned a p-value matrix into long form with `expand_grid(x = directs, y = indirects)`, a direct port of the deposit's `expand.grid` call. Base `expand.grid` varies its first argument fastest and tidyr's `expand_grid` varies its last, and the matrix is read column-major, so the two axes were swapped and every p-value map this repository drew was the transpose of its own estimates. No estimate was affected: point estimates and confidence regions come from the fit object rather than from the plot, which is why cell-by-cell agreement with the archive's matrices passed throughout. It surfaced only when a map was laid beside the one the corrigendum prints. The argument order and the reason for it are now recorded in `helpers.R`.

The test statistic goes through `.lm.fit` rather than `lm`. The rewrite fits about eleven million regressions; `lm` takes 0.295 milliseconds each and `.lm.fit` takes 0.0035, and the two agree on the sum of squared residuals to floating point. That is the difference between a five-minute run and an hour-long one, and a reproduction check nobody runs is not a check.

7.2 Deprecated patterns replaced

Table 7: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd(””)‘ and ‘Replication Archive/‘ paths	‘here::here()‘
‘maptools::readShapeSpatial‘	‘sf::st_read‘ + ‘spdep::poly2nb‘
‘lattice::levelplot‘ + ‘larrows‘	‘ggplot2::geom_tile‘ + ‘annotate(”segment”)‘

‘pdf()‘ / ‘dev.off()‘	‘ggsave()‘, PDF and PNG
‘xtable‘ + ‘sink()‘ to a hardcoded path	‘write_csv()‘ to ‘output/‘
‘save()‘ / ‘load()‘ of .rdata	‘write_rds()‘ / ‘read_rds()‘
nested ‘for‘ loops over ‘data.frame(...)‘ + ‘lm()‘	design matrices + ‘.lm.fit‘
‘Z_block[, 1]‘ inside the simulation loop	‘z_block[, sample(ncol(z_block), 1)]‘

7.3 Summary tables

Table 8: Appendix Table 1, constant effects models, as the maintained rewrite produc uncorrected column is the analysis the article published; the column beside same analysis on the ideal points the corrigendum used.

	No spillover	Absolute distance	Absolute distance (uncorrected)	Absolute distance (corrigendum's original panel)	F
Direct Effect	-0.16	-0.25	-0.3	-0.3	-
Direct 95% CI	[-0.34,0.02]	[-0.625,NA]	[-0.5,-0.075]	[-0.575,0]	[-
Indirect Effect		-0.15	-0.225	-0.175	-
Indirect 95% CI		[-0.35,0.175]	[-0.325,-0.1]	[-0.325,-0.025]	[-
Corresponding Figure	1a	2	2	2	A

Table 9: Appendix Table 2, heterogeneous effects models, as the maintained rewrite produces it.

	No spillover	Absolute distance	Absolute distance (uncorrected)
Direct Effect, High Support	0.05	-0.125	-0.1875
Direct 95% CI, High Support	[-0.225,0.35]	[-0.65,NA]	[-0.4,0.075]
Indirect Effect, High Support		-0.225	-0.3
Indirect 95% CI, High Support		[-0.45,NA]	[-0.425,-0.175]
Direct Effect, Low Support	-0.375	-0.425	-0.475
Direct 95% CI, Low Support	[-0.675,-0.05]	[-0.775,0.05]	[-0.8,-0.125]
Indirect Effect, Low Support		-0.1	-0.175
Indirect 95% CI, Low Support		[-0.325,0.15]	[-0.325,0]
Corresponding Figure	1b	3	3

7.4 In-text quantities

Table 10: The page 167 arithmetic, under both similarity matrices.

Quantity	Value
Legislators treated	35
Difference-in-means estimate (Figure 1a)	-0.16
Implied change in votes under the difference in means	-5.6
Legislators voting yea	51
Yea votes in the uniformity trial, uncorrected	61.46
Total causal effect in votes, uncorrected	10.46
Yea votes in the uniformity trial, corrected	59.64

The article states that 61 legislators would have voted yea in the uniformity trial against 51 who actually did, for a total causal effect of 10 votes, roughly twice the 5.6 the difference in means implies. The uncorrected column reproduces that. The corrected column is the quantity the corrigendum's conclusion bears on: the point estimates barely move, so the arithmetic barely moves, and what changes is that neither effect can be distinguished from zero.

8 Figure verification

The correction's visible consequence is the width of the shaded region. The uncorrected map is a tight blob well away from both axes; the corrected map spreads across zero on both. The third map is the same uncorrected analysis on the corrigendum's ideal points, and it sits between them: still short of zero on the indirect axis, but reaching it on the direct one. The corrigendum prints two panels, and they are the first and third of these; the second is the article's own Figure 2, which the corrigendum does not reprint.

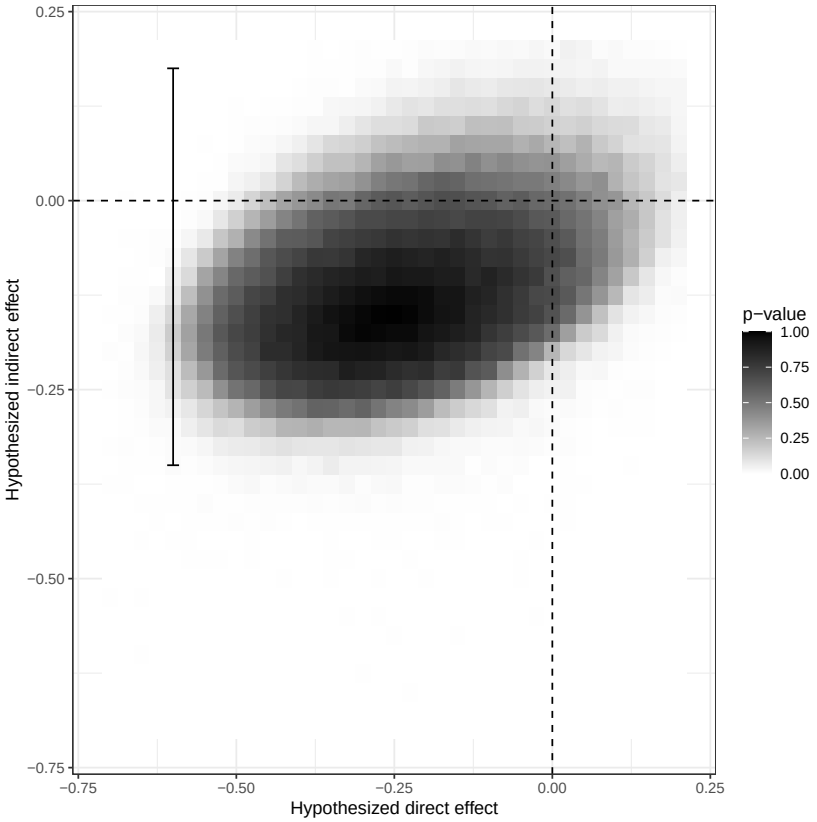


Figure 1: Figure 2, corrected similarity matrix.

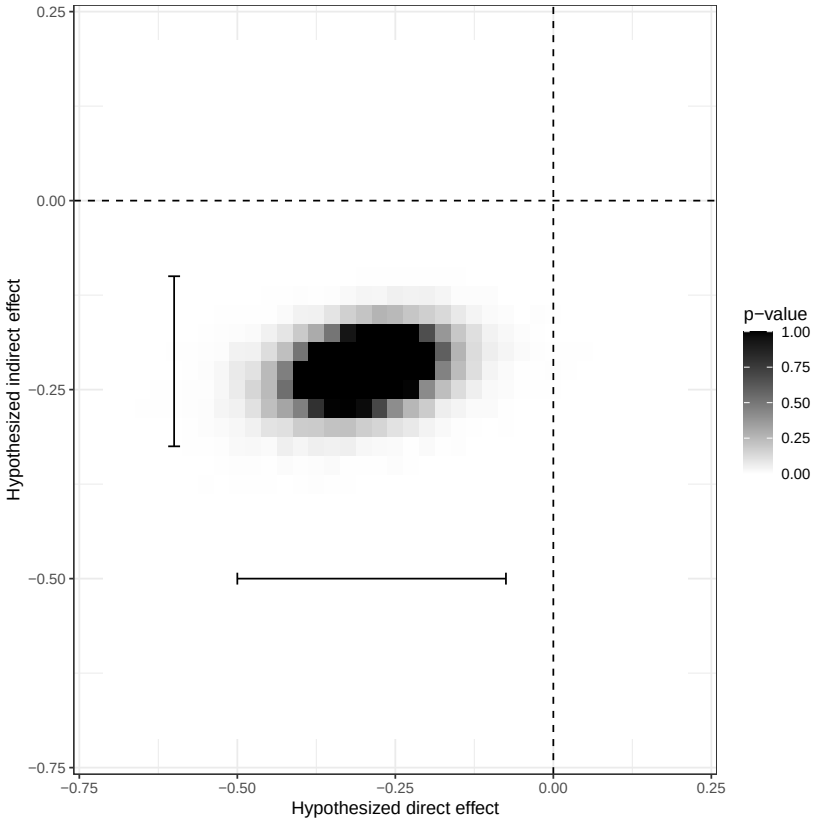


Figure 2: Figure 2 as published, all-ones similarity matrix on the deposited ideal points.

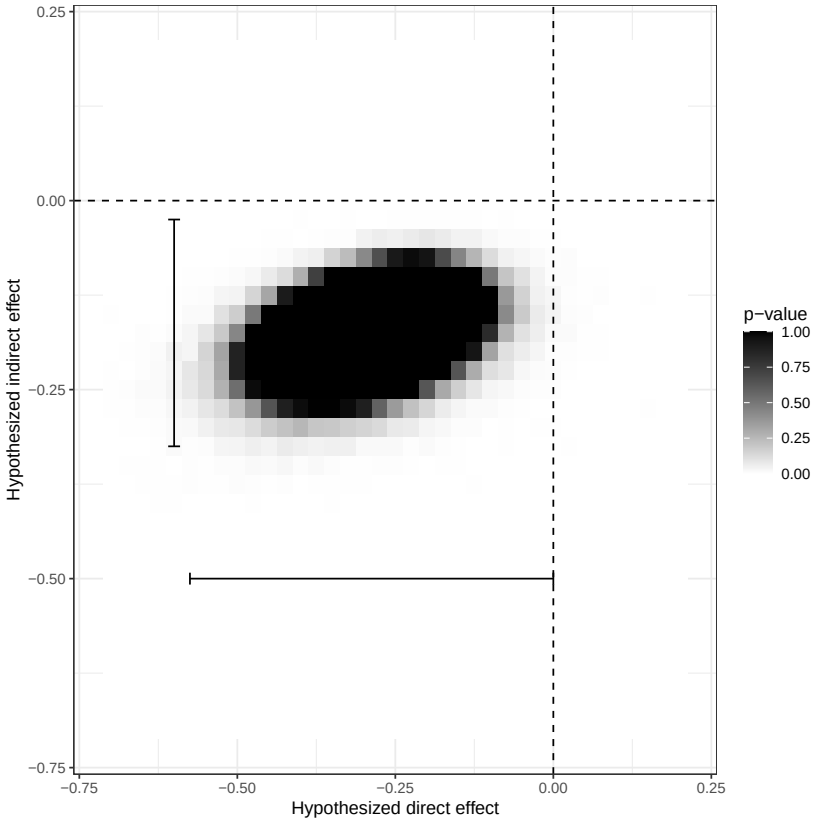


Figure 3: The corrigendum's own original panel: the same all-ones matrix on the ideal points the corrigendum was computed from. The wider plateau is why its restated intervals are wider than the article's.

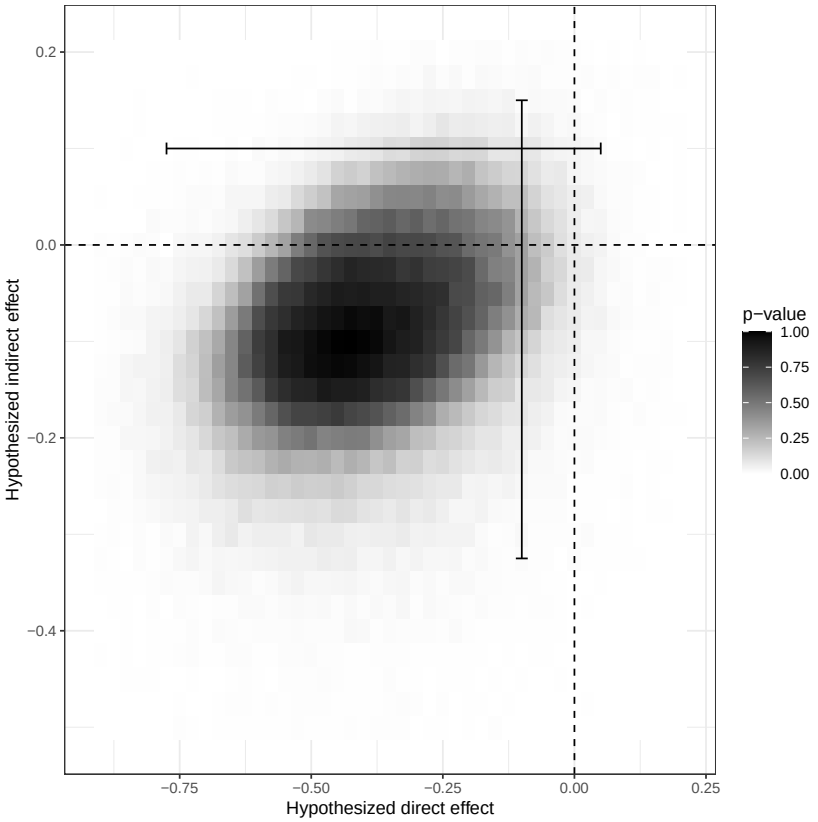


Figure 4: Figure 3a, lower-support districts, corrected.

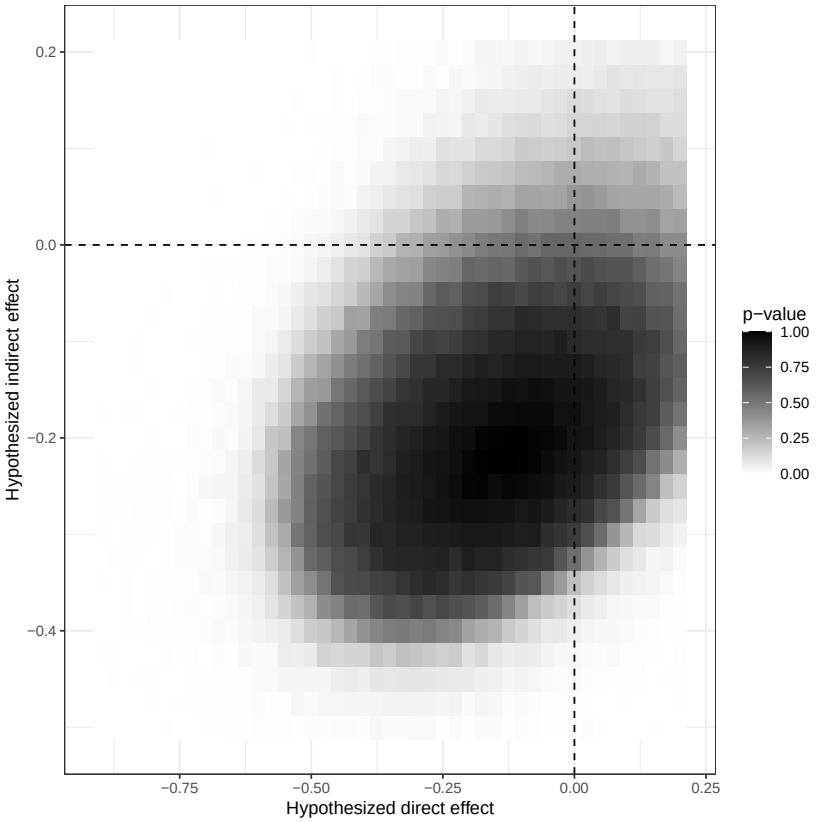


Figure 5: Figure 3b, higher-support districts, corrected.

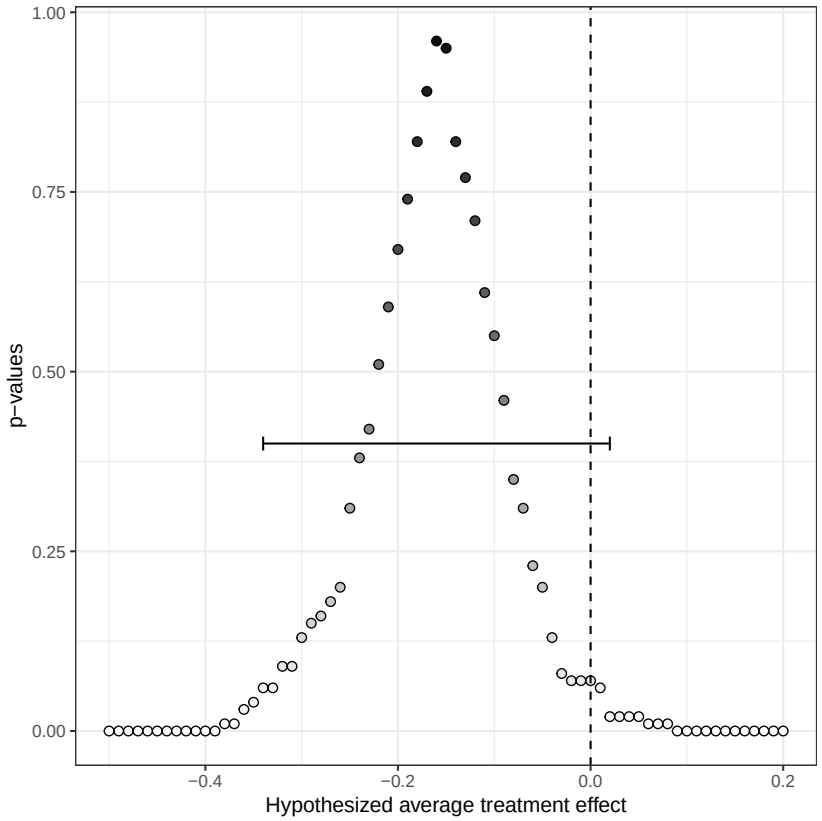


Figure 6: Figure 1a, the no-spillover constant effects model, which reproduces exactly.

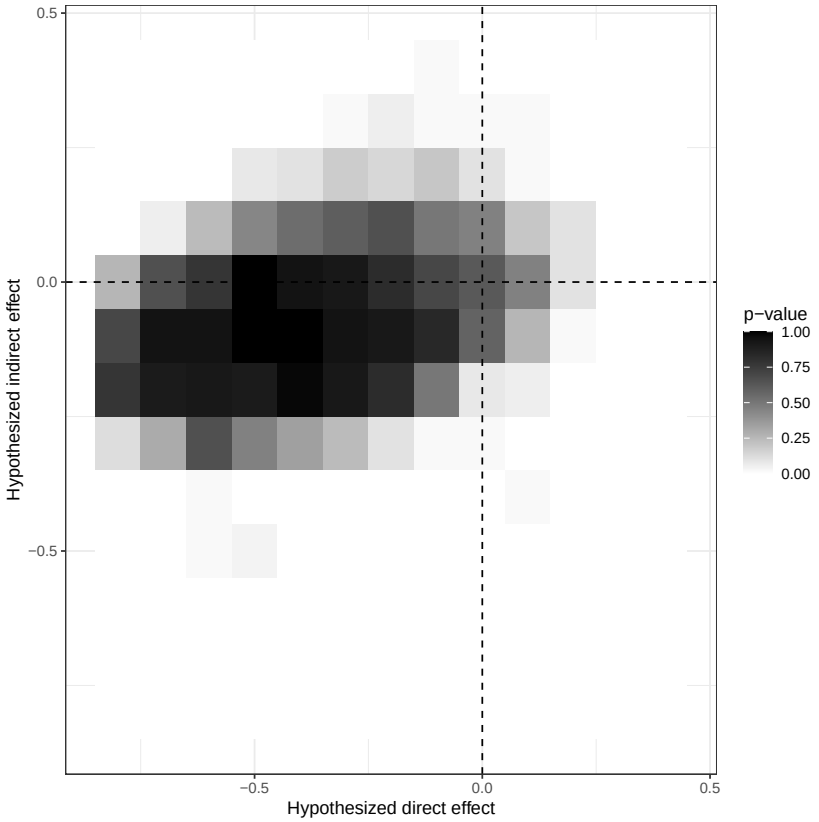


Figure 7: Appendix Figure 5a with a fresh assignment drawn per simulation.

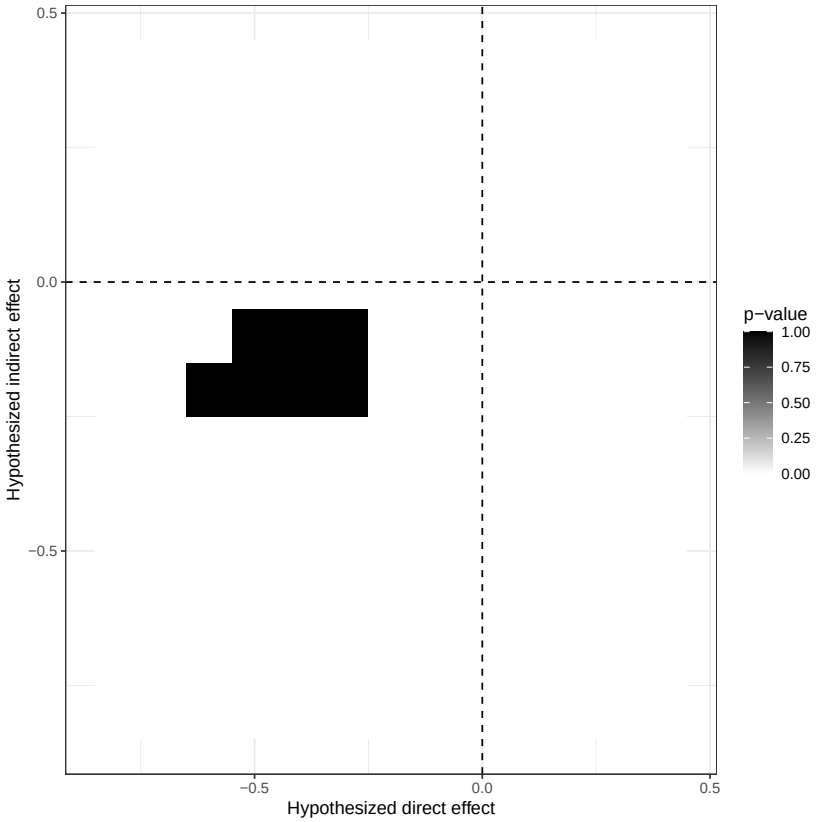


Figure 8: Appendix Figure 5a as published: every p-value is 0 or 1.

9 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-03

Table 12: Package versions used for the run behind this report.

Package	Version
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2

purrr	1.2.2
readr	2.2.0
sf	1.1.2
spdep	1.4.2
here	1.0.2